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Revision History				
Marketing PN	Rev.	Date	By	Description
X-NAVQ95-IO	X1	2026-05-29	JvE (CTO SI)	Initial schematic development
X-NAVQ95-IO	A	2026-06-17	JvE (CTO SI)	Schematic is released
NAVQ95-IO	A1	2026-10-09	JvE (CTO SI)	Schematic is released to production



NOTES:

1. Interrupted lines coded with the same letter or letter combinations are electrically connected.
2. Device type number is for reference only. The number varies with the manufacturer.
3. Greyed out components indicate Do Not Populate (DNP) during assembly.
4. Special signal usage:
 - _B denotes an active-low signal.
 - _P and _N denote a differential signal.
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.
6. All SILK display properties will be in this color
7. All schematic annotations will be in this color

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Designer: Jari van Ewijk	Drawing Title: NAVQ95-IO		
Drawn by: Jari van Ewijk	Page Title: Cover		
Approved: Cristin Oudrei	Size C	Document Number SCH-97013 / PDF: SPF-97013	Rev A1
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DNP Components

REF DES	ASSY_OPT	PAGE NAME
R21,R29,R48	DNP	04 - I/O EXP & CAN TRX
R60	DNP	05 - Sensors & Connectors
C9,C11,C48,R93,R96	DNP	06 - Audio
R2,R6,R69,R74,R77	DNP	07 - Console

Switch Default Positions

REF DES	SWITCH	PAGE NAME
S1	OPEN	05 - Sensors & Connectors



Classification: Public

Drawing Title:

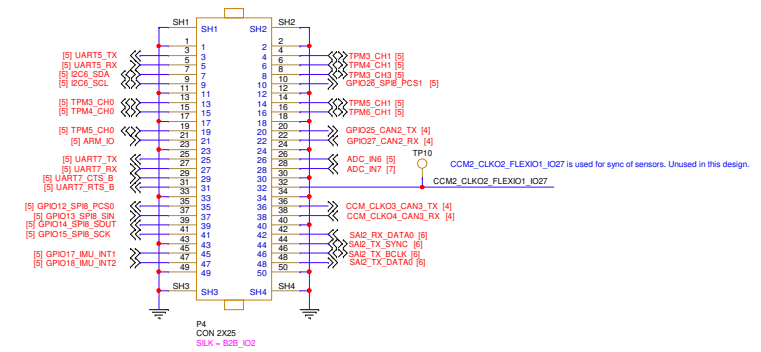
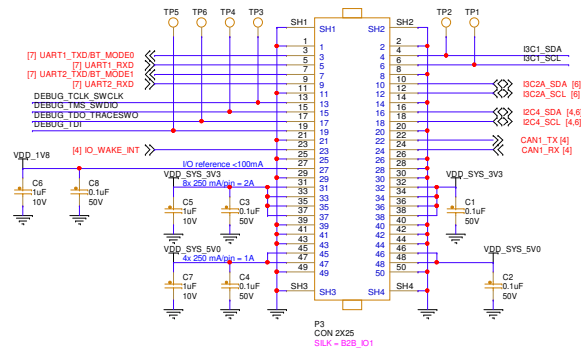
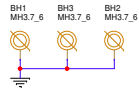
NAVQ95-IO

Page Title:

Tables

Size	Document Number	Rev
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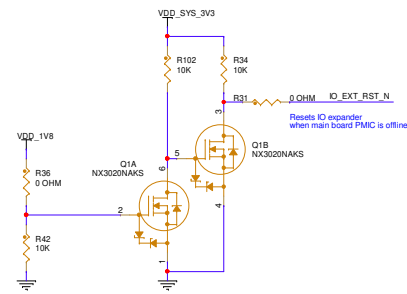
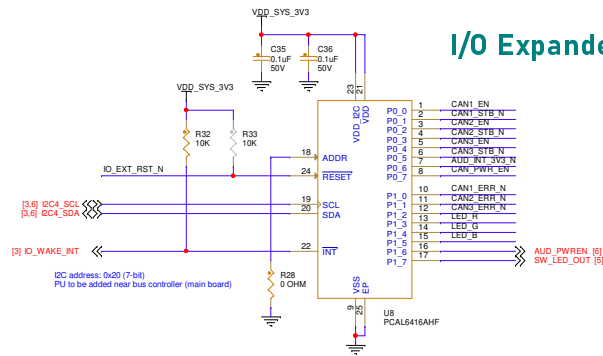
B2B Interface



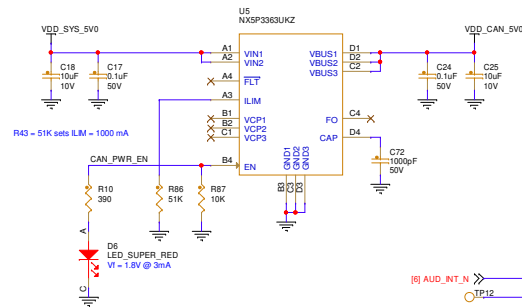
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Drawing Title: NAVQ95-IO			
Page Title: B2B Interface			
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I/O EXP & CAN TRX

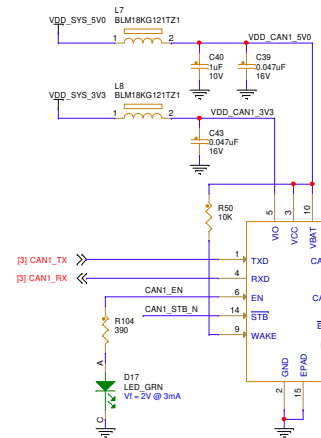
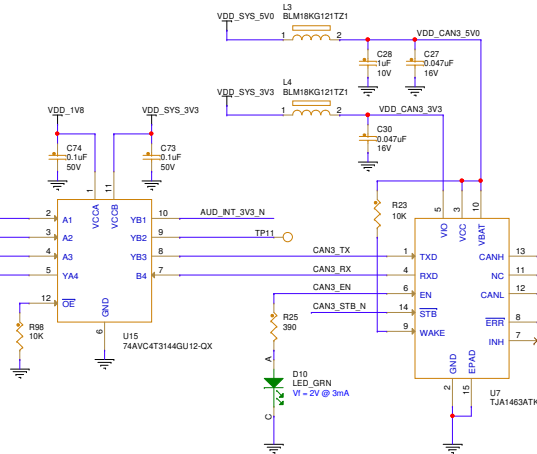
I/O Expander



CAN Bus Current Limiting

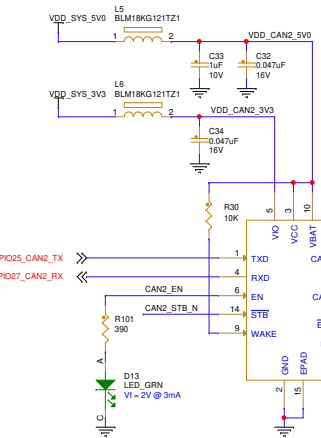


[5] AUD_INT_N
[5] CCM_CLK03_CAN3_TX
[5] CCM_CLK04_CAN3_RX



CAN1

CAN: 120 ohm



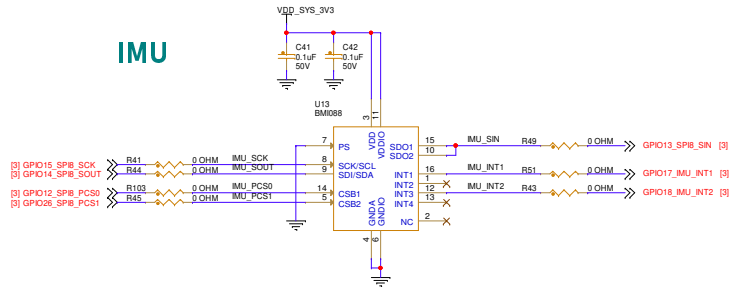
CAN2

CAN3

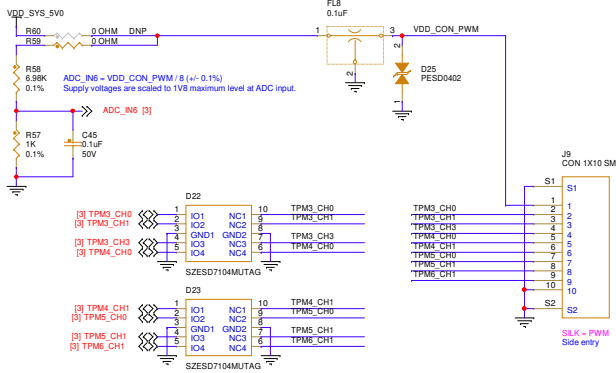
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Sensors & Connectors

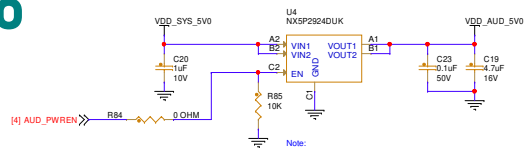
IMU



Be careful when putting more than 5V on VDD_CON_PWM!
R50 should not be populated when providing external power!
ADC_IN6 is scaled to handle maximum 12V

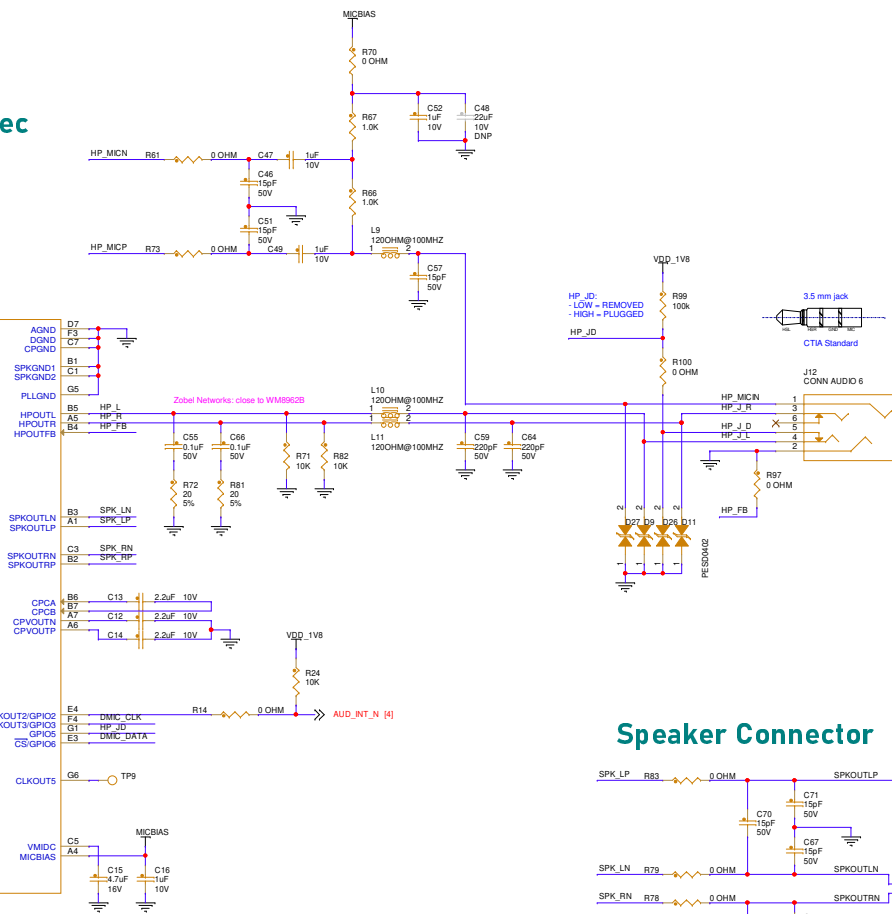
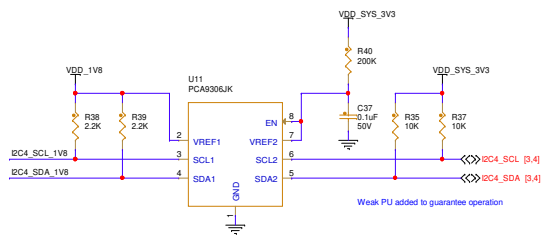
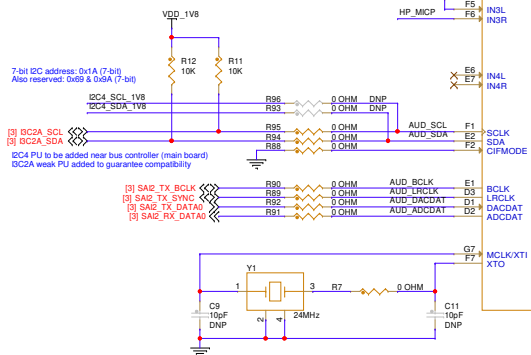
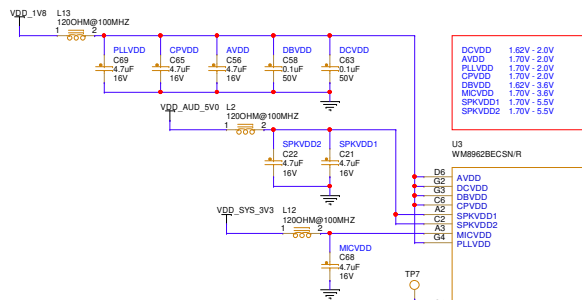


Audio

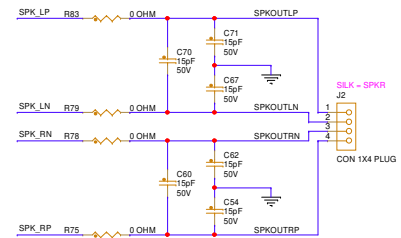


Note:
1. Default low to reduce pop noise
2. High enable for speaker output.

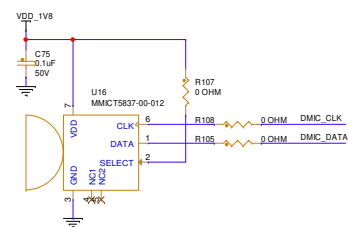
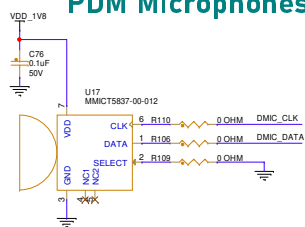
Audio Codec



Speaker Connector



PDM Microphones



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Audio

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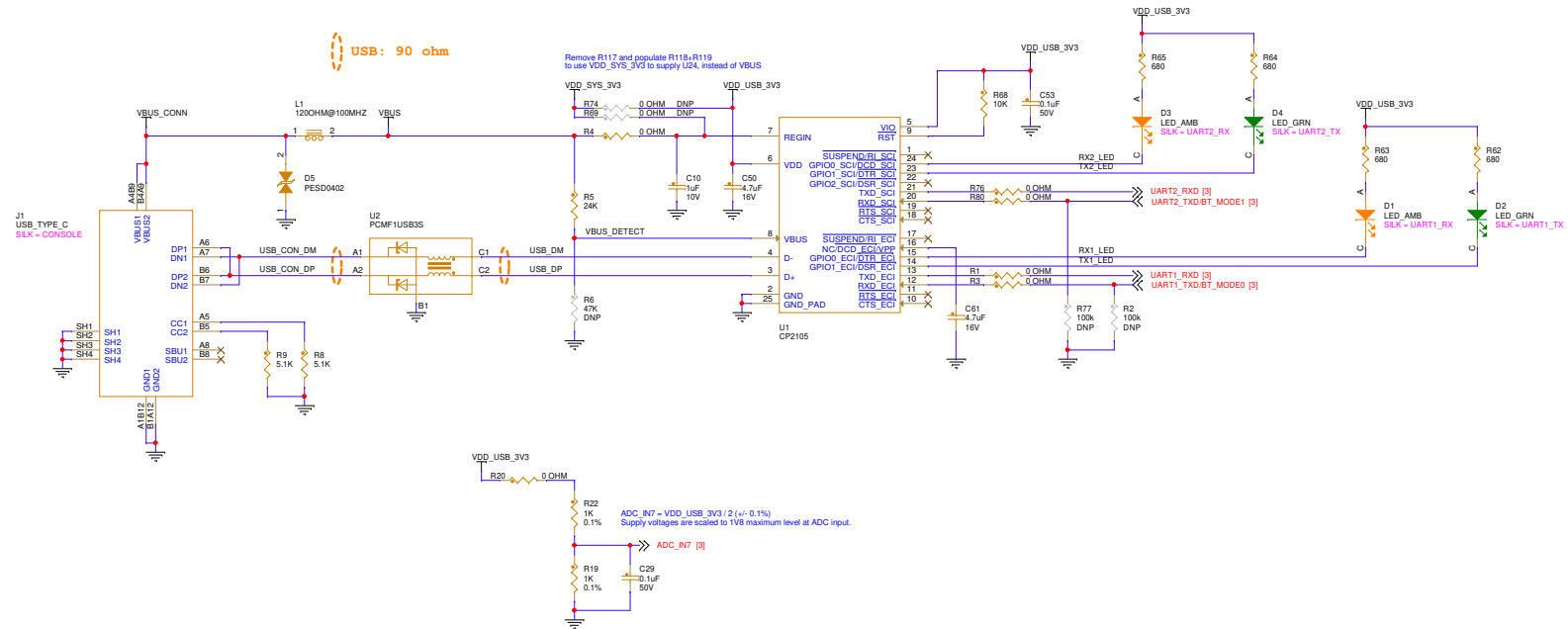
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Sheet 6

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Direct	9
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Console



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